



UNIVERSIDADE FEDERAL DO RIO GRANDE DO NORTE
DEPARTAMENTO DE CIÊNCIAS ATMOSFÉRICAS E CLIMÁTICAS

CURSO: METEOROLOGIA
CLIMATOLOGIA ESTOCASTICA II
PROFESSOR: PAULO SÉRGIO LÚCIO

SEXTO PERÍODO

UNIDADE 1

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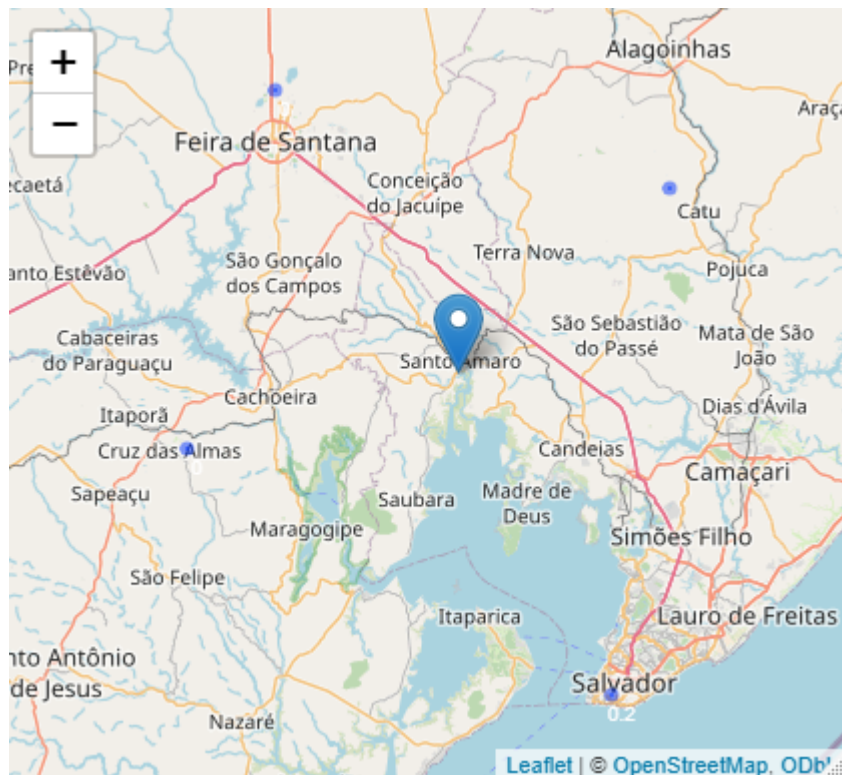


ATIVIDADE

NATAL - RN
2023

O presente trabalho terá como finalidade a probabilidade de ter uma previsão por utilização de dados disponíveis de cidades próximas, utilizando a interpolação de dados feito por Italo Cegatta para a cidade de Santo Amaro - BA, para prevenir a precipitação. As cidades que foram tomadas com base foram Salvador, Feira de Santana e Cruz das Almas.

```
> if (!require("pacman")) install.packages("pacman")
> pacman::p_load(readr, dplyr, leaflet)
> # importa o arquivo os dados de chuva
> santo.amaro <- read.csv("C:/Users/vitoria/Downloads/santo amaro - Página1.csv")
Warning message:
In read.table(file = file, header = header, sep = sep, quote = quote, :
  incomplete final line found by readTableHeader on 'C:/Users/vitoria/Downloads/santo amaro -
  Página1.csv'
> dados <- santo.amaro
> dados
  cidade lat lon alt p
1 salvador -13.00 -38.50 47.70 0.2
2 feira -12.19 -38.96 229.60 0.0
3 cruz -12.67 -39.08 219.76 0.0
4 santo -12.32 -38.42 42.00 NA
> leaflet(dados) %>%
+ addTiles() %>%
+ addMarkers(-38.71, -12.57) %>%
+ addCircleMarkers(
+   ~lon, ~lat,
+   radius = ~p * 0.8,
+   label = ~as.character(p),
+   popup = ~cidade,
+   fillOpacity = 0.6,
+   labelOptions = labelOptions(
+     style = list("color" = "white"),
+     offset = c(5, -10),
+     noHide = TRUE,
+     textOnly = TRUE,
+     direction = "bottom"
+   )
+ )
```



```

+ )
> haversine <- function(lon1, lat1, lon2, lat2) {
+ # converte graus pra radiano
+ rad <- pi/180
+ # raio medio da terra no equador em km
+ R <- 6378.1
+
+ dlon <- (lon2 - lon1) * rad
+ dlat <- (lat2 - lat1) * rad
+
+ a <- (sin(dlat/2))^2 +
+   cos(lat1 * rad) *
+   cos(lat2 * rad) *
+   (sin(dlon/2))^2
+
+ c <- 2 * atan2(sqrt(a), sqrt(1 - a))
+
+ d <- R * c
+
+ # distancia em km
+ return(d)
+ }
> dist <- dados %>%
+   mutate("d_sta" = haversine(lon, lat, -38.71, -12.57))
> dist
  cidade lat lon alt p d_sta

```

```

1 salvador -13.00 -38.50 47.70 0.2 53.01865
2 feira -12.19 -38.96 229.60 0.0 50.28199
3 cruz -12.67 -39.08 219.76 0.0 41.70596
4 santo -12.32 -38.42 42.00 NA 42.05056
> idw <- function(x, dist, na.rm = TRUE) {
+ s1 <- sum(x / dist^2, na.rm = na.rm)
+ s2 <- sum(1 / dist^2, na.rm = na.rm)
+
+ return(s1 / s2)
+ }
> "dados_sta" <- dist %>%
+ add_row(
+   .,
+   cidade = "Santo",
+   lon = -38.71,
+   lat = -12.57,
+   p = round(idw(.$p, .$d_sta), 1)
+ )
> dados_sta
  cidade lat lon alt p d_sta
1 salvador -13.00 -38.50 47.70 0.2 53.01865
2 feira -12.19 -38.96 229.60 0.0 50.28199
3 cruz -12.67 -39.08 219.76 0.0 41.70596
4 santo -12.32 -38.42 42.00 NA 42.05056
5 Santo -12.57 -38.71 NA 0.0 NA

```

Destinado para outro tipo denominado “Estima - T”, necessitando para estimar os dados de temperatura, média, máxima e mínima. Porém com a última atualização não fica disponível em sua latitude até -10°. Assim, a sua precisão acaba diminuindo ainda mais sua porcentagem.

Área 3: Alagoas, Sergipe e Bahia

Localidade: Santo Amaro

Longitude: -38,71°

Latitude: -10°

Altitude: 8m

Temperaturas (°C) Média, Máxima e Mínima

Ano	Mês	T-Med	T-Max	T-Min
1950	01	28,81	33,27	22,43
1950	02	29,05	33,63	22,78
1950	03	29,47	33,80	23,53
1950	04	28,12	32,19	22,93
1950	05	26,45	30,35	22,43
1950	06	25,18	29,55	20,71
1950	07	24,21	28,40	19,79
1950	08	24,66	28,55	19,12
1950	09	25,89	30,82	19,58
1950	10	28,28	32,79	21,45
1950	11	28,92	33,86	22,52
1950	12	28,22	32,90	22,27
1951	01	28,88	33,34	22,50
1951	02	29,23	33,81	22,96
1951	03	29,40	33,73	23,46
1951	04	28,28	32,35	23,09
1951	05	26,59	30,49	22,57
1951	06	25,60	29,97	21,13
1951	07	24,58	28,77	20,16
1951	08	25,33	29,22	19,79
1951	09	26,54	31,47	20,23
1951	10	29,00	33,51	22,17
1951	11	29,75	34,69	23,35
1951	12	28,86	33,54	22,91
1952	01	29,37	33,83	22,99
1952	02	29,72	34,30	23,45
1952	03	29,84	34,17	23,90
1952	04	28,57	32,64	23,38
1952	05	26,78	30,68	22,76
1952	06	25,40	29,77	20,93
1952	07	24,21	28,40	19,79
1952	08	24,98	28,87	19,44
1952	09	26,24	31,17	19,93
1952	10	28,60	33,11	21,77
1952	11	29,24	34,18	22,84
1952	12	28,48	33,16	22,53

1953	01	29,28	33,74	22,90
1953	02	29,62	34,20	23,35
1953	03	29,83	34,16	23,89
1953	04	28,62	32,69	23,43
1953	05	26,87	30,77	22,85
1953	06	25,58	29,95	21,11
1953	07	24,56	28,75	20,14
1953	08	25,17	29,06	19,63
1953	09	26,50	31,43	20,19
1953	10	28,66	33,17	21,83
1953	11	29,48	34,42	23,08
1953	12	28,60	33,28	22,65
1954	01	29,20	33,66	22,82
1954	02	29,57	34,15	23,30
1954	03	29,70	34,03	23,76
1954	04	28,20	32,27	23,01
1954	05	26,56	30,46	22,54
1954	06	25,13	29,50	20,66
1954	07	23,95	28,14	19,53
1954	08	24,73	28,62	19,19
1954	09	25,97	30,90	19,66
1954	10	28,32	32,83	21,49
1954	11	28,97	33,91	22,57
1954	12	28,18	32,86	22,23
1955	01	28,75	33,21	22,37
1955	02	29,18	33,76	22,91
1955	03	29,41	33,74	23,47
1955	04	28,09	32,16	22,90
1955	05	26,29	30,19	22,27
1955	06	25,03	29,40	20,56
1955	07	24,06	28,25	19,64
1955	08	24,73	28,62	19,19
1955	09	25,97	30,90	19,66
1955	10	28,04	32,55	21,21
1955	11	28,76	33,70	22,36
1955	12	28,10	32,78	22,15
1956	01	28,69	33,15	22,31
1956	02	29,19	33,77	22,92
1956	03	29,35	33,68	23,41
1956	04	28,13	32,20	22,94
1956	05	26,49	30,39	22,47
1956	06	25,18	29,55	20,71
1956	07	24,01	28,20	19,59
1956	08	24,77	28,66	19,23
1956	09	25,98	30,91	19,67
1956	10	28,37	32,88	21,54
1956	11	29,10	34,04	22,70
1956	12	28,39	33,07	22,44

1957	01	28,95	33,41	22,57
1957	02	29,58	34,16	23,31
1957	03	29,81	34,14	23,87
1957	04	28,67	32,74	23,48
1957	05	27,12	31,02	23,10
1957	06	25,75	30,12	21,28
1957	07	24,66	28,85	20,24
1957	08	25,40	29,29	19,86
1957	09	26,47	31,40	20,16
1957	10	28,80	33,31	21,97
1957	11	29,77	34,71	23,37
1957	12	29,05	33,73	23,10
1958	01	29,63	34,09	23,25
1958	02	30,08	34,66	23,81
1958	03	30,14	34,47	24,20
1958	04	28,78	32,85	23,59
1958	05	26,91	30,81	22,89
1958	06	25,56	29,93	21,09
1958	07	24,53	28,72	20,11
1958	08	25,16	29,05	19,62
1958	09	26,32	31,25	20,01
1958	10	28,70	33,21	21,87
1958	11	29,40	34,34	23,00
1958	12	28,86	33,54	22,91
1959	01	29,39	33,85	23,01
1959	02	29,73	34,31	23,46
1959	03	29,81	34,14	23,87
1959	04	28,63	32,70	23,44
1959	05	26,97	30,87	22,95
1959	06	25,50	29,87	21,03
1959	07	24,29	28,48	19,87
1959	08	24,96	28,85	19,42
1959	09	26,29	31,22	19,98
1959	10	28,71	33,22	21,88
1959	11	29,37	34,31	22,97
1959	12	28,58	33,26	22,63
1960	01	29,10	33,56	22,72
1960	02	29,38	33,96	23,11
1960	03	29,78	34,11	23,84
1960	04	28,47	32,54	23,28
1960	05	26,74	30,64	22,72
1960	06	25,34	29,71	20,87
1960	07	24,41	28,60	19,99
1960	08	25,01	28,90	19,47
1960	09	26,41	31,34	20,10
1960	10	28,60	33,11	21,77
1960	11	29,29	34,23	22,89
1960	12	28,71	33,39	22,76

1961	01	29,20	33,66	22,82
1961	02	29,68	34,26	23,41
1961	03	29,71	34,04	23,77
1961	04	28,54	32,61	23,35
1961	05	26,82	30,72	22,80
1961	06	25,56	29,93	21,09
1961	07	24,28	28,47	19,86
1961	08	24,96	28,85	19,42
1961	09	26,16	31,09	19,85
1961	10	28,52	33,03	21,69
1961	11	29,32	34,26	22,92
1961	12	28,61	33,29	22,66
1962	01	29,17	33,63	22,79
1962	02	29,50	34,08	23,23
1962	03	29,65	33,98	23,71
1962	04	28,41	32,48	23,22
1962	05	26,74	30,64	22,72
1962	06	25,55	29,92	21,08
1962	07	24,48	28,67	20,06
1962	08	25,18	29,07	19,64
1962	09	26,39	31,32	20,08
1962	10	28,71	33,22	21,88
1962	11	29,47	34,41	23,07
1962	12	28,60	33,28	22,65
1963	01	29,13	33,59	22,75
1963	02	29,54	34,12	23,27
1963	03	29,78	34,11	23,84
1963	04	28,59	32,66	23,40
1963	05	26,92	30,82	22,90
1963	06	25,70	30,07	21,23
1963	07	24,87	29,06	20,45
1963	08	25,50	29,39	19,96
1963	09	26,67	31,60	20,36
1963	10	29,05	33,56	22,22
1963	11	29,83	34,77	23,43
1963	12	29,01	33,69	23,06
1964	01	29,53	33,99	23,15
1964	02	29,71	34,29	23,44
1964	03	29,74	34,07	23,80
1964	04	28,44	32,51	23,25
1964	05	26,52	30,42	22,50
1964	06	25,20	29,57	20,73
1964	07	24,24	28,43	19,82
1964	08	24,71	28,60	19,17
1964	09	25,89	30,82	19,58
1964	10	28,31	32,82	21,48
1964	11	28,90	33,84	22,50
1964	12	28,06	32,74	22,11

1965	01	28,75	33,21	22,37
1965	02	29,32	33,90	23,05
1965	03	29,54	33,87	23,60
1965	04	28,40	32,47	23,21
1965	05	26,85	30,75	22,83
1965	06	25,56	29,93	21,09
1965	07	24,54	28,73	20,12
1965	08	25,22	29,11	19,68
1965	09	26,55	31,48	20,24
1965	10	28,96	33,47	22,13
1965	11	29,72	34,66	23,32
1965	12	28,99	33,67	23,04
1966	01	29,51	33,97	23,13
1966	02	29,83	34,41	23,56
1966	03	29,97	34,30	24,03
1966	04	28,62	32,69	23,43
1966	05	26,81	30,71	22,79
1966	06	25,62	29,99	21,15
1966	07	24,59	28,78	20,17
1966	08	25,10	28,99	19,56
1966	09	26,38	31,31	20,07
1966	10	28,69	33,20	21,86
1966	11	29,39	34,33	22,99
1966	12	28,62	33,30	22,67
1967	01	29,04	33,50	22,66
1967	02	29,43	34,01	23,16
1967	03	29,60	33,93	23,66
1967	04	28,40	32,47	23,21
1967	05	26,80	30,70	22,78
1967	06	25,46	29,83	20,99
1967	07	24,24	28,43	19,82
1967	08	24,85	28,74	19,31
1967	09	26,09	31,02	19,78
1967	10	28,43	32,94	21,60
1967	11	29,12	34,06	22,72
1967	12	28,43	33,11	22,48
1968	01	28,88	33,34	22,50
1968	02	29,29	33,87	23,02
1968	03	29,47	33,80	23,53
1968	04	28,32	32,39	23,13
1968	05	26,60	30,50	22,58
1968	06	25,56	29,93	21,09
1968	07	24,59	28,78	20,17
1968	08	25,26	29,15	19,72
1968	09	26,52	31,45	20,21
1968	10	28,93	33,44	22,10
1968	11	29,67	34,61	23,27
1968	12	28,96	33,64	23,01

1969	01	29,68	34,14	23,30
1969	02	30,05	34,63	23,78
1969	03	30,22	34,55	24,28
1969	04	28,95	33,02	23,76
1969	05	27,34	31,24	23,32
1969	06	25,81	30,18	21,34
1969	07	24,62	28,81	20,20
1969	08	25,31	29,20	19,77
1969	09	26,61	31,54	20,30
1969	10	29,04	33,55	22,21
1969	11	29,71	34,65	23,31
1969	12	29,04	33,72	23,09
1970	01	29,63	34,09	23,25
1970	02	29,85	34,43	23,58
1970	03	29,94	34,27	24,00
1970	04	28,68	32,75	23,49
1970	05	26,86	30,76	22,84
1970	06	25,37	29,74	20,90
1970	07	24,24	28,43	19,82
1970	08	24,82	28,71	19,28
1970	09	26,12	31,05	19,81
1970	10	28,41	32,92	21,58
1970	11	29,08	34,02	22,68
1970	12	28,13	32,81	22,18
1971	01	28,77	33,23	22,39
1971	02	29,07	33,65	22,80
1971	03	29,33	33,66	23,39
1971	04	28,26	32,33	23,07
1971	05	26,58	30,48	22,56
1971	06	25,29	29,66	20,82
1971	07	24,25	28,44	19,83
1971	08	24,84	28,73	19,30
1971	09	26,06	30,99	19,75
1971	10	28,35	32,86	21,52
1971	11	29,03	33,97	22,63
1971	12	28,21	32,89	22,26
1972	01	29,02	33,48	22,64
1972	02	29,58	34,16	23,31
1972	03	29,66	33,99	23,72
1972	04	28,62	32,69	23,43
1972	05	27,04	30,94	23,02
1972	06	25,82	30,19	21,35
1972	07	24,88	29,07	20,46
1972	08	25,59	29,48	20,05
1972	09	26,79	31,72	20,48
1972	10	29,33	33,84	22,50
1972	11	30,16	35,10	23,76
1972	12	29,44	34,12	23,49

1973	01	29,89	34,35	23,51
1973	02	30,11	34,69	23,84
1973	03	30,06	34,39	24,12
1973	04	28,64	32,71	23,45
1973	05	26,86	30,76	22,84
1973	06	25,42	29,79	20,95
1973	07	24,39	28,58	19,97
1973	08	24,94	28,83	19,40
1973	09	26,21	31,14	19,90
1973	10	28,44	32,95	21,61
1973	11	29,10	34,04	22,70
1973	12	28,16	32,84	22,21
1974	01	28,57	33,03	22,19
1974	02	29,03	33,61	22,76
1974	03	29,35	33,68	23,41
1974	04	28,11	32,18	22,92
1974	05	26,51	30,41	22,49
1974	06	25,34	29,71	20,87
1974	07	24,37	28,56	19,95
1974	08	24,99	28,88	19,45
1974	09	26,12	31,05	19,81
1974	10	28,39	32,90	21,56
1974	11	29,10	34,04	22,70
1974	12	28,32	33,00	22,37
1975	01	28,88	33,34	22,50
1975	02	29,29	33,87	23,02
1975	03	29,42	33,75	23,48
1975	04	28,19	32,26	23,00
1975	05	26,45	30,35	22,43
1975	06	25,11	29,48	20,64
1975	07	24,18	28,37	19,76
1975	08	24,70	28,59	19,16
1975	09	25,83	30,76	19,52
1975	10	28,07	32,58	21,24
1975	11	28,82	33,76	22,42
1975	12	27,90	32,58	21,95
1976	01	28,51	32,97	22,13
1976	02	29,11	33,69	22,84
1976	03	29,36	33,69	23,42
1976	04	28,15	32,22	22,96
1976	05	26,52	30,42	22,50
1976	06	25,41	29,78	20,94
1976	07	24,36	28,55	19,94
1976	08	25,02	28,91	19,48
1976	09	26,40	31,33	20,09
1976	10	28,83	33,34	22,00
1976	11	29,59	34,53	23,19
1976	12	28,73	33,41	22,78

1977	01	29,32	33,78	22,94
1977	02	29,57	34,15	23,30
1977	03	29,73	34,06	23,79
1977	04	28,41	32,48	23,22
1977	05	26,87	30,77	22,85
1977	06	25,64	30,01	21,17
1977	07	24,51	28,70	20,09
1977	08	25,11	29,00	19,57
1977	09	26,51	31,44	20,20
1977	10	28,87	33,38	22,04
1977	11	29,65	34,59	23,25
1977	12	28,89	33,57	22,94
1978	01	29,42	33,88	23,04
1978	02	29,66	34,24	23,39
1978	03	29,82	34,15	23,88
1978	04	28,40	32,47	23,21
1978	05	26,69	30,59	22,67
1978	06	25,39	29,76	20,92
1978	07	24,42	28,61	20,00
1978	08	24,99	28,88	19,45
1978	09	26,28	31,21	19,97
1978	10	28,60	33,11	21,77
1978	11	29,42	34,36	23,02
1978	12	28,74	33,42	22,79
1979	01	29,37	33,83	22,99
1979	02	29,69	34,27	23,42
1979	03	29,94	34,27	24,00
1979	04	28,67	32,74	23,48
1979	05	26,98	30,88	22,96
1979	06	25,81	30,18	21,34
1979	07	24,58	28,77	20,16
1979	08	25,19	29,08	19,65
1979	09	26,65	31,58	20,34
1979	10	28,90	33,41	22,07
1979	11	29,67	34,61	23,27
1979	12	28,94	33,62	22,99
1980	01	29,53	33,99	23,15
1980	02	29,80	34,38	23,53
1980	03	29,94	34,27	24,00
1980	04	28,70	32,77	23,51
1980	05	27,08	30,98	23,06
1980	06	25,77	30,14	21,30
1980	07	24,57	28,76	20,15
1980	08	25,13	29,02	19,59
1980	09	26,55	31,48	20,24
1980	10	28,79	33,30	21,96
1980	11	29,62	34,56	23,22
1980	12	28,81	33,49	22,86

1981	01	29,21	33,67	22,83
1981	02	29,47	34,05	23,20
1981	03	29,83	34,16	23,89
1981	04	28,45	32,52	23,26
1981	05	26,84	30,74	22,82
1981	06	25,59	29,96	21,12
1981	07	24,50	28,69	20,08
1981	08	25,02	28,91	19,48
1981	09	26,50	31,43	20,19
1981	10	28,94	33,45	22,11
1981	11	29,65	34,59	23,25
1981	12	28,94	33,62	22,99
1982	01	29,49	33,95	23,11
1982	02	29,79	34,37	23,52
1982	03	29,94	34,27	24,00
1982	04	28,78	32,85	23,59
1982	05	27,23	31,13	23,21
1982	06	25,95	30,32	21,48
1982	07	24,74	28,93	20,32
1982	08	25,42	29,31	19,88
1982	09	26,80	31,73	20,49
1982	10	29,30	33,81	22,47
1982	11	30,13	35,07	23,73
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1983	01	30,10	34,56	23,72
1983	02	30,41	34,99	24,14
1983	03	30,50	34,83	24,56
1983	04	29,13	33,20	23,94
1983	05	27,43	31,33	23,41
1983	06	26,20	30,57	21,73
1983	07	24,98	29,17	20,56
1983	08	25,55	29,44	20,01
1983	09	26,69	31,62	20,38
1983	10	28,84	33,35	22,01
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1983	12	28,79	33,47	22,84
1984	01	29,25	33,71	22,87
1984	02	29,60	34,18	23,33
1984	03	29,88	34,21	23,94
1984	04	28,55	32,62	23,36
1984	05	26,73	30,63	22,71
1984	06	25,41	29,78	20,94
1984	07	24,48	28,67	20,06
1984	08	25,14	29,03	19,60
1984	09	26,58	31,51	20,27
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1984	11	29,39	34,33	22,99
1984	12	28,46	33,14	22,51

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1985	02	29,46	34,04	23,19
1985	03	29,62	33,95	23,68
1985	04	28,41	32,48	23,22
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1985	07	24,43	28,62	20,01
1985	08	25,08	28,97	19,54
1985	09	26,38	31,31	20,07
1985	10	28,69	33,20	21,86
1985	11	29,50	34,44	23,10
1985	12	28,73	33,41	22,78
1986	01	29,16	33,62	22,78
1986	02	29,67	34,25	23,40
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1986	05	26,94	30,84	22,92
1986	06	25,73	30,10	21,26
1986	07	24,67	28,86	20,25
1986	08	25,31	29,20	19,77
1986	09	26,68	31,61	20,37
1986	10	29,07	33,58	22,24
1986	11	29,87	34,81	23,47
1986	12	29,09	33,77	23,14
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1987	06	26,23	30,60	21,76
1987	07	25,18	29,37	20,76
1987	08	25,86	29,75	20,32
1987	09	27,21	32,14	20,90
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1991	05	27,35	31,25	23,33
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1992	07	24,73	28,92	20,31
1992	08	25,21	29,10	19,67
1992	09	26,53	31,46	20,22
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1993	03	29,98	34,31	24,04
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1993	05	27,28	31,18	23,26
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1993	11	29,87	34,81	23,47
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1995	01	29,76	34,22	23,38
1995	02	30,09	34,67	23,82
1995	03	30,20	34,53	24,26
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1995	05	27,17	31,07	23,15
1995	06	25,97	30,34	21,50
1995	07	24,88	29,07	20,46
1995	08	25,36	29,25	19,82
1995	09	26,58	31,51	20,27
1995	10	28,86	33,37	22,03
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1995	12	28,77	33,45	22,82
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1996	09	26,63	31,56	20,32
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1997	02	29,63	34,21	23,36
1997	03	29,95	34,28	24,01
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1997	07	25,19	29,38	20,77
1997	08	25,89	29,78	20,35
1997	09	27,23	32,16	20,92
1997	10	29,69	34,20	22,86
1997	11	30,55	35,49	24,15
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1998	01	30,42	34,88	24,04
1998	02	30,75	35,33	24,48
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1998	08	25,61	29,50	20,07
1998	09	26,81	31,74	20,50
1998	10	29,02	33,53	22,19
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2000	07	24,58	28,77	20,16
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2002	06	26,10	30,47	21,63
2002	07	24,99	29,18	20,57
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2004	02	30,09	34,67	23,82
2004	03	30,17	34,50	24,23
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2004	07	24,75	28,94	20,33
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2005	02	30,11	34,69	23,84
2005	03	30,33	34,66	24,39
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2005	05	27,33	31,23	23,31
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2006	05	27,24	31,14	23,22
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2008	09	26,80	31,73	20,49
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2010	03	30,59	34,92	24,65
2010	04	29,34	33,41	24,15
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2011	05	26,91	30,81	22,89
2011	06	25,66	30,03	21,19
2011	07	24,62	28,81	20,20
2011	08	25,22	29,11	19,68
2011	09	26,39	31,32	20,08
2011	10	28,71	33,22	21,88
2011	11	29,42	34,36	23,02
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2014	08	25,56	29,45	20,02
2014	09	26,82	31,75	20,51
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2014	11	29,94	34,88	23,54
2014	12	29,14	33,82	23,19
2015	01	29,58	34,04	23,20
2015	02	29,94	34,52	23,67
2015	03	30,13	34,46	24,19
2015	04	29,01	33,08	23,82
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2015	12	29,86	34,54	23,91
2016	01	30,36	34,82	23,98
2016	02	30,50	35,08	24,23
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2016	04	29,29	33,36	24,10
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2016	06	26,12	30,49	21,65
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2016	08	25,51	29,40	19,97
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2017	02	30,00	34,58	23,73
2017	03	30,21	34,54	24,27
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2017	09	26,68	31,61	20,37
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2018	12	29,26	33,94	23,31
2019	01	29,54	34,00	23,16
2019	02	29,95	34,53	23,68
2019	03	30,20	34,53	24,26
2019	04	28,93	33,00	23,74
2019	05	27,23	31,13	23,21
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2019	07	24,83	29,02	20,41
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2020	09	26,48	31,41	20,17
2020	10	28,73	33,24	21,90
2020	11	29,44	34,38	23,04
2020	12	28,65	33,33	22,70

2021	01	29,22	33,68	22,84
2021	02	29,63	34,21	23,36
2021	03	29,88	34,21	23,94
2021	04	28,60	32,67	23,41
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2021	11	29,70	34,64	23,30
2021	12	28,74	33,42	22,79
2022	01	29,37	33,83	22,99
2022	02	29,65	34,23	23,38
2022	03	29,85	34,18	23,91
2022	04	28,58	32,65	23,39
2022	05	26,89	30,79	22,87
2022	06	25,71	30,08	21,24
2022	07	24,56	28,75	20,14
2022	08	25,23	29,12	19,69
2022	09	26,49	31,42	20,18
2022	10	28,73	33,24	21,90
2022	11	29,49	34,43	23,09

Utilizando o método Knn - vizinhos:

```
> # Carregue a biblioteca 'class'
> library(class)
> # Crie dados hipotéticos ideais
> set.seed(0) # Defina uma semente para reprodutibilidade
> # Gere dados de treinamento
> X_treinamento <- seq(1, 10, by = 0.5) # Recursos (features)
> y_treinamento <- seq(10, 100, by = 5) # Precipitação (valores ideais)
> # Gere dados de teste
> X_teste <- seq(1.5, 10.5, by = 0.5) # Recursos para teste
> y_teste <- seq(15, 105, by = 5) # Precipitação correspondente (valores ideais)
> # Escolha um valor ideal para 'k' com base no MSE obtido (53)
> k <- 1
> # Treine o modelo k-NN
> modelo_knn <- knn(train = data.frame(X = X_treinamento), test = data.frame(X = X_teste), cl
= y_treinamento, k = k)
> # Avalie o modelo
> erro_quadratico_medio <- sqrt(mean((as.numeric(modelo_knn) - y_teste)^2))
> cat("Erro Quadrático Médio (RMSE) com dados ideais:", erro_quadratico_medio, "\n")
Erro Quadrático Médio (RMSE) com dados ideais: 53.75872
> library(class)
> # Número de vizinhos (k) para o k-NN
> k <- 1
```

```

> # Dados de treinamento (use um subconjunto dos dados 'fortaleza' como exemplo)
> dados1 <- data.frame(temp = fortaleza$temp[1:2])
> # Dados de teste (um único ponto de teste)
> novo_ponto <- data.frame(temp = 27.7) # Substitua 25 pela temperatura do ponto que você
deseja prever
> # Realize a previsão com k-NN
> previsao <- knn(train = dados1, test = novo_ponto, k = k)
Error in knn(train = dados1, test = novo_ponto, k = k) :
  no missing values are allowed
> # Imprima a previsão
> cat("Previsão de temp para o novo ponto:", previsao, "\n")
Error in cat("Previsão de temp para o novo ponto:", previsao, "\n") :
  object 'previsao' not found

```

Houve a tentativa de realizar a previsão, mas o “erro” não conseguiu ser corrigido.